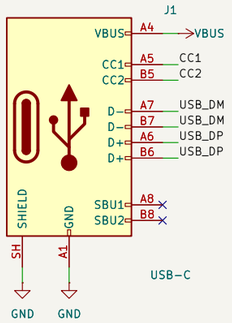
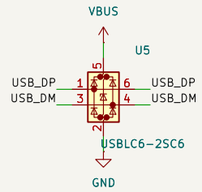
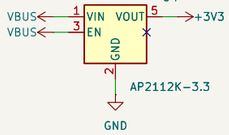
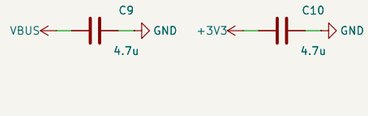
USB-C (sink, USB 2.0 FS) + ESD + 3.3 V LDO

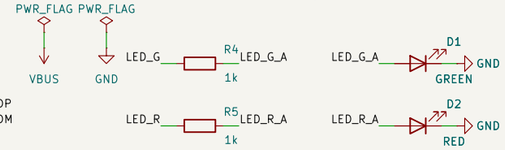


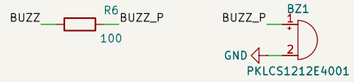


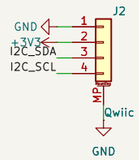


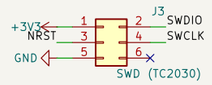


User interface / connectors

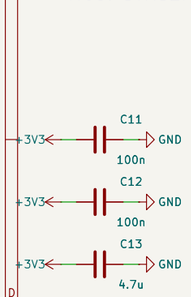


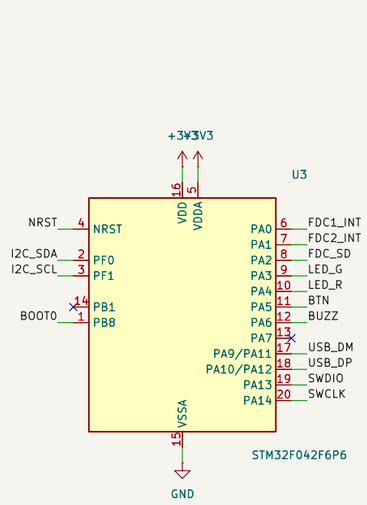


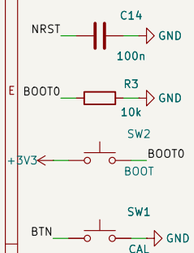




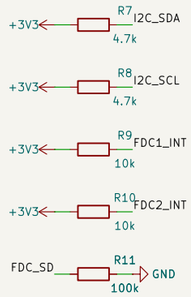
MCU: STM32F042F6P6 (crystal-less USB, ROM DFU via BOOT0)



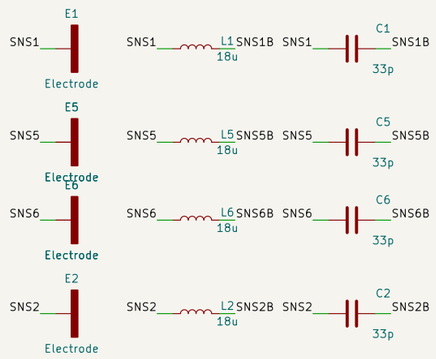


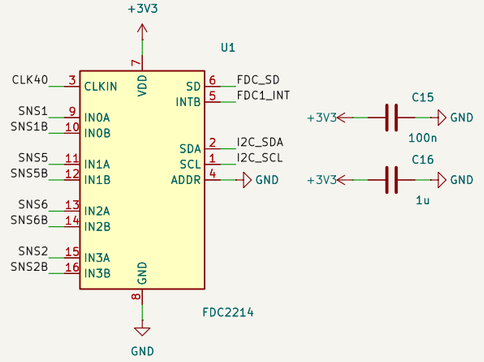


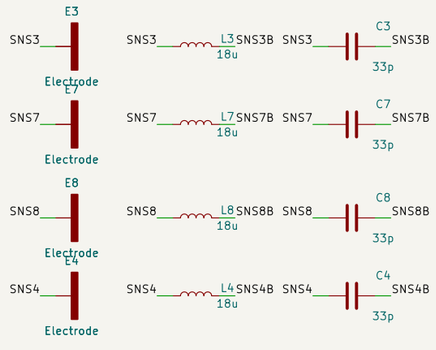
Bus pull-ups

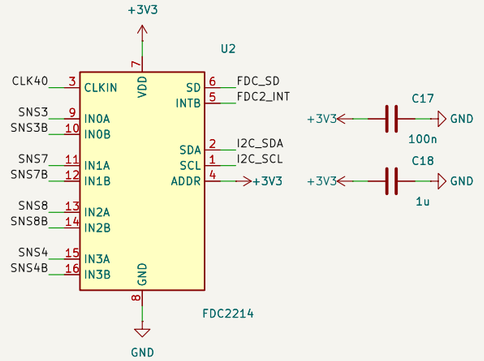


Capacitance-to-digital: 2x FDC2214, single-ended sensors, 18uH || 33pF tanks

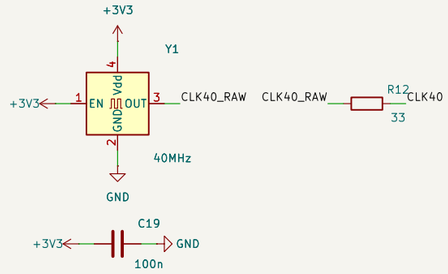








40 MHz reference (shared)



Near-field H-loop at the tip, for an external VNA (NanoVNA etc.)

50 ohm microstrip on F.Cu over In1 GND (w = 0.35 mm, JLC04161H-7628)



H1 M2

H2 M2

CONCEPT rev 0.1 — not yet fabricated or validated. See README.md.

https://github.com/KarpelesLab/ledger_implant_detector

CONCEPT DESIGN — NOT YET BUILT OR VALIDATED

Sheet: /

File: implant_detector.kicad_sch

Title: Ledger Implant Detector (CONCEPT)

Size: A3	Date: 2026-10-18	**Rev: 0.1**
KiCad E.D.A. 10.0.3		Id: 1/1